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Milestone 4

Database Design Overview

For this milestone, I enhanced the Daily Drop Android app by redesigning its database to use the Room Persistence Library. The previous version relied on a manual SQLite helper that required direct SQL queries, which made maintenance harder and increased the risk of errors. By integrating Room, the app now manages data through clearly defined entities, data access objects (DAOs), and a repository layer. This new design uses a `WeightEntity` class to store each record with an ID, date, and weight value. The Room database provides better structure, automatic schema handling, and improved reliability while following the Model-View-ViewModel pattern for clean separation between the data and the user interface.

Enhancements Implemented

The main enhancement was migrating from manual SQL to Room's annotation-based operations. This eliminated the need for raw SQL strings and improved code readability. I created a `WeightRepositoryRoom` class to manage all insert, delete, and query operations on background threads, ensuring smooth performance without freezing the interface. The `HomeActivity` now interacts directly with the repository, updating the `RecyclerView` list instantly whenever data changes. I also improved the Add Weight dialog by pre-filling the date, validating inputs, and formatting values to display one decimal point followed by "lbs." These updates make the app look cleaner and operate more professionally.

Impact on Functionality

This database enhancement made the app faster, safer, and easier to maintain. Using Room removed the risk of SQL syntax errors and ensured that all schema changes are managed automatically. Running operations on background threads improved responsiveness, allowing users to save or delete entries instantly without lag. The new structure also supports scalability, making it easier to add future features like graphs, analytics, or new metrics. Overall, the app now feels more polished and reliable, reflecting real-world software engineering standards.

Future Considerations

Moving forward, I plan to implement LiveData and ViewModel components so the app can update the interface automatically when the database changes. I also want to add cloud backup using Firebase so users can save their progress online. Another improvement will involve encrypting data and securing user information for long-term privacy. These updates will make the app more dynamic, secure, and ready for future growth.

Reflection

Completing this milestone strengthened my understanding of database architecture and Android development. Migrating to Room taught me how professional apps handle persistence, concurrency, and data safety. I also gained experience designing clean, scalable code through the repository pattern. This project gave me confidence in my ability to build real-world, data-driven applications that are stable, efficient, and maintainable.